

COOLSHADE · REQUIREMENTS

CoolShade requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-25	Amish Chadha	Initial scaffold

CoolShade requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with users, and will be checked by calculation at TRL 3 and revised after co-design sessions (see CSH-PRB-001). Two requirements are not met by the current concept (R13 cost and, in some conditions, R4 cooling), and three are at risk (R5 wetting, R9 water hygiene, R16 scaling).

Design day: 38 °C air, 25 % relative humidity, clear sky, wind about 1 m/s, 5 h of misting demand, 4.5 peak sun hours. **Humid check:** 35 °C and 60 % relative humidity.

Table 1. Requirements, targets and status of the TRL 2 concept (status values are estimates).

ID	Requirement	Target	Verification (TRL 3 or later)	Concept status
R1	Shaded waiting area	6 m ² or more of shade at solar noon; room for at least 6 people standing or seated and one wheelchair space of 0.8 x 1.2 m	Plan layout; sun path check per site	Met by design: 7.2 m ² roof
R2	Headroom	Clear height 2.2 m or more under any part of the frame, nozzle or brace in the waiting area	Model check	Met by design: about 2.1 m under knee braces at the rear posts, which sit outside the main waiting area; about 2.5 m or more elsewhere. Brace position to confirm
R3	Radiant heat reduction	Mean radiant temperature under the canopy at least 15 °C below full sun at midday on the design day	Literature comparison; later field measurement with a globe thermometer (HeatMap Node method)	Expected met: about 17.3 °C measured for shade sails in Tempe (Middel et al., 2021); unverified for this design
R4	Air cooling from misting	Air temperature in the waiting zone (0.5 to 2.0 m height) at least 2 °C below ambient during misting on the design day	Energy balance at TRL 3; later field test	Not met at 1 m/s wind: about 1.4 °C averaged over the duty cycle, about 2.7 °C while spraying (estimate). Met in calmer air
R5	No wetting	No perceptible wetting of skin or clothing at 1.5 m height on the design day; no puddles on the waiting area	Droplet size data from nozzle maker; later field test	At risk: low-pressure nozzles make larger droplets than high-pressure systems
R6	Smart activation	Mists only when air temperature is 32 °C or more, relative humidity is 60 % or less, presence is detected in the last 2 min and the tank is above its low level; stops within 60 s when any condition fails	Control logic review; later bench test	Met by design (thresholds proposed, awaiting Amish)
R7	Water use	100 L or less per design day	Nozzle flow calculation	Met: about 85 L per design day (estimate)
R8	Energy autonomy	Energy-neutral on the design day from the onboard panel; battery covers one full design day with no sun	Energy budget	Met: about 205 Wh per day used against about 340 Wh harvested; about 205 Wh usable storage (estimates)
R9	Water hygiene	Potable-quality supply water; 5 µm filtration; mist line drained within 5 min after every session; water in the tank 48 h old or less; opaque, shaded, lockable tank; documented weekly disinfection	Design review; later microbiological sampling with a partner lab	At risk: stored water on hot days will sit in the 20 to 50 °C range in which <i>Legionella</i> grows (WHO); relies on turnover, draining and disinfection
R10	Wind resistance	Frame and anchors survive a 3 s gust of 30 m/s (108 km/h) with the fabric	Structural calculation at TRL 3	Unverified: first estimate about 144 MPa bending stress in each

ID	Requirement	Target	Verification (TRL 3 or later)	Concept status
		fitted; fabric removable in 15 min or less before forecast storms		post as a cantilever, before the benefit of knee braces
R11	Privacy	No camera or microphone; presence by passive infrared only; if a status link is fitted, only counts and levels (on-time, water used, temperature, humidity, faults) leave the device	Design review	Met by design
R12	Electrical safety	12 V DC only, no mains; battery fused at the terminal; all electronics in a lockable IP65 enclosure; LiFePO4 chemistry with BMS and cold-charge cutoff	Design review	Met by design
R13	Cost	Parts for one prototype \$700 or less	BOM	Not met: about \$735 (indicative), about 5 % over
R14	Buildability	Bolted assembly; no welding; built with a drill, angle grinder, spanners and a ladder by two people in 2 days or less	Assembly review	Met by design; build time unverified
R15	Refill interval	1 design day or more between tank refills	Water balance	Met: 120 L tank covers about 1.4 design days (estimate)
R16	Maintenance	Nozzles cleanable or replaceable without tools in 15 min or less; filter changed in 10 min or less; descaling interval 1 month or longer	Design review; later field log	At risk in hard-water areas: scale can block 0.4 mm orifices
R17	Movability	Can be unbolted and moved to another site by two people in 1 day or less; no part heavier than 40 kg to lift (tank empty)	Mass estimate	Met: heaviest part a post of about 20 kg (estimate)

■ Assumptions

- Nozzle flow about 4 L/h each at about 7 bar, eight nozzles, 50 % spray duty cycle while active. To be checked against nozzle datasheets.
- About 87 % of the sprayed water evaporates in the air; the rest drifts away, drips or wets surfaces (estimate).
- Cooling is estimated over a control volume 3.0 m long and 2.0 m high with air crossing it at the wind speed. Real mixing is less even than this.
- Panel output 100 W x 4.5 peak sun hours x 0.75 system factor.